RADIATOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5]

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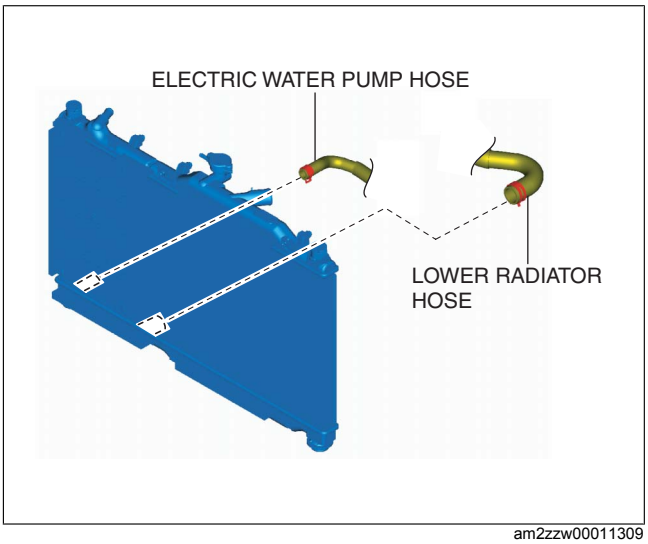
Warning

- **If the cooling system cap or the water-cooled charge air cooler reserve tank cap is removed when the engine is hot, steam and boiling water may be injected, causing severe burns or severe injury. Perform the removal of the cooling system cap and water-cooled charge air cooler reserve tank cap when the engine is cool.**
- **Turn off the engine and wait until it is cool. Even then, be very careful when removing the cap. Wrap a thick cloth around it and slowly turn it counterclockwise to the first stop. Step back while the pressure escapes.**
- **When you are sure all the pressure is gone, press down on the cap using the cloth, turn it, and remove it.**

Note

- Remove/install the radiator with the shroud panel set aside temporarily.

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/ CONNECTION.)
2. Remove the engine cover. (See ENGINE COVER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
3. Remove the front under cover No.1. (See FRONT UNDER COVER No.1 REMOVAL/INSTALLATION.)
4. Drain the engine coolant and water-cooled charge air cooler coolant. (See ENGINE COOLANT REPLACEMENT [SKYACTIV-D 1.5].)(See WATER-COOLED CHARGE AIR COOLER COOLANT REPLACEMENT [SKYACTIV-D 1.5].)
5. Disconnect the lower radiator hose and the electric water pump hose from the radiator.
6. Remove the air cleaner and fresh air duct component. (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
7. Remove the following parts so that the shroud panel can be set aside temporarily.
   - Front bumper (See FRONT BUMPER REMOVAL.)(See FRONT BUMPER INSTALLATION.)
   - Front combination light (See FRONT COMBINATION LIGHT REMOVAL/ INSTALLATION.)
   - Bumper stiffener lower (See BUMPER STIFFENER LOWER REMOVAL/ INSTALLATION.)
   - Seal plate (See SEAL PLATE REMOVAL/ INSTALLATION.)
   - Front bumper reinforcement (See FRONT BUMPER REINFORCEMENT REMOVAL/INSTALLATION.)



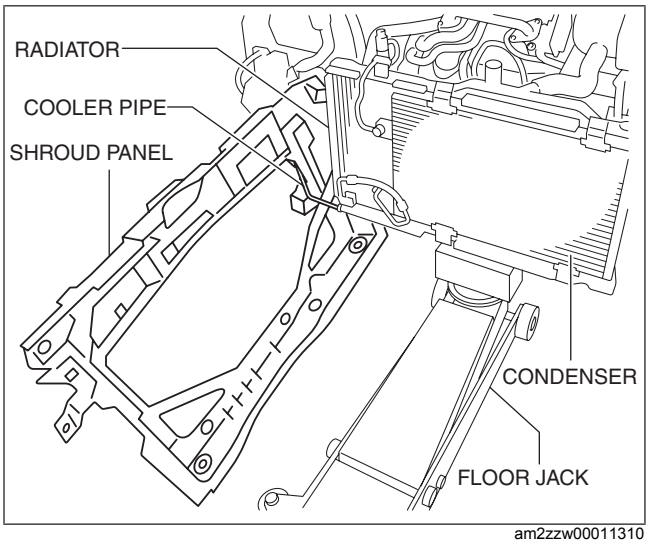
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8. Set the shroud panel aside with the cooler pipe connected as shown in the figure. (See SHROUD PANEL REMOVAL/INSTALLATION.)

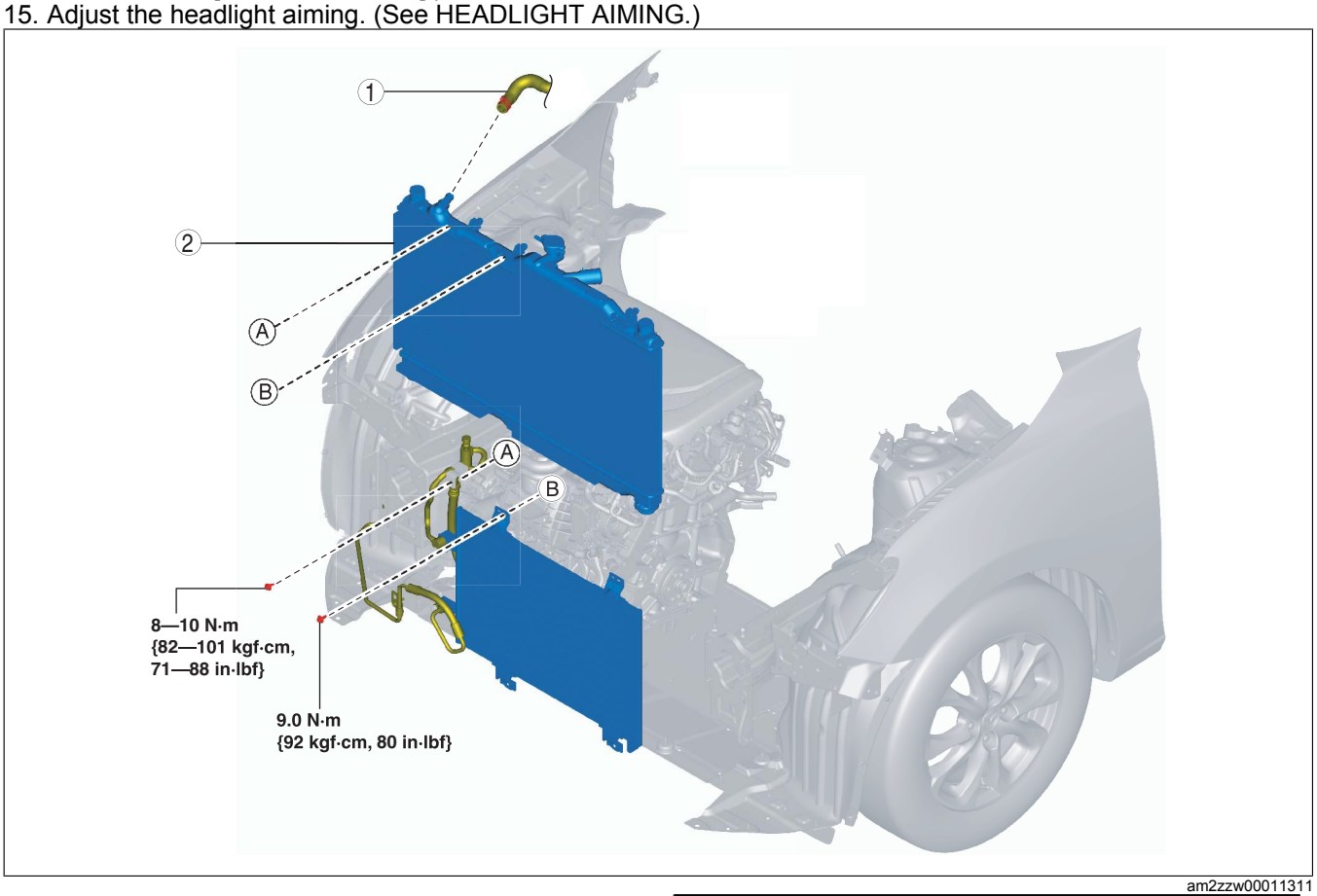
Note

- Support the radiator using a floor jack.

9. Remove the coolant reserve tank. (See COOLANT RESERVE TANK REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
10. Remove the cooling fan component. (See COOLING FAN MOTOR REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)
11. Remove in the order indicated in the table.
12. Install in the reverse order of removal.
13. Refill the engine coolant and water-cooled charge air cooler coolant. (See ENGINE COOLANT REPLACEMENT [SKYACTIV-D 1.5].)(See WATER-COOLED CHARGE AIR COOLER COOLANT REPLACEMENT [SKYACTIV-D 1.5].)
14. Inspect for engine coolant leakage and water-cooled charge air cooler coolant leakage. (See ENGINE COOLANT LEAKAGE INSPECTION [SKYACTIV-D 1.5].)(See WATER-COOLED CHARGE AIR COOLER COOLANT LEAKAGE INSPECTION [SKYACTIV-D 1.5].)
15. Adjust the headlight aiming. (See HEADLIGHT AIMING.)



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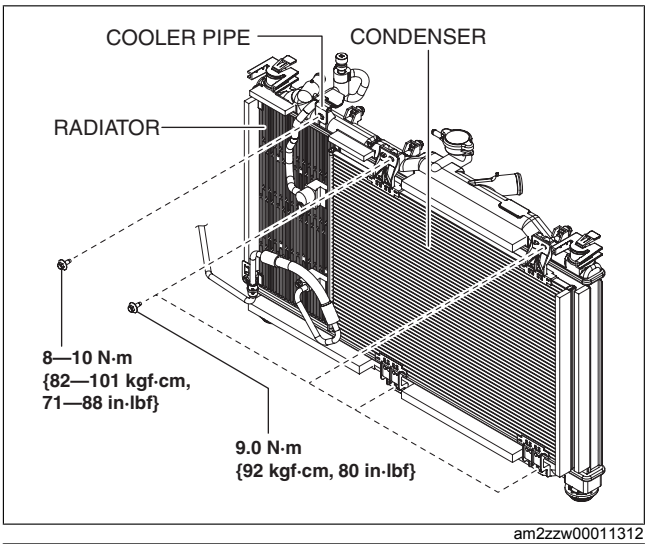
1	Water-cooled charge air cooler reserve tank hose

2	Radiator (See Radiator Removal Note.)

Radiator Removal Note

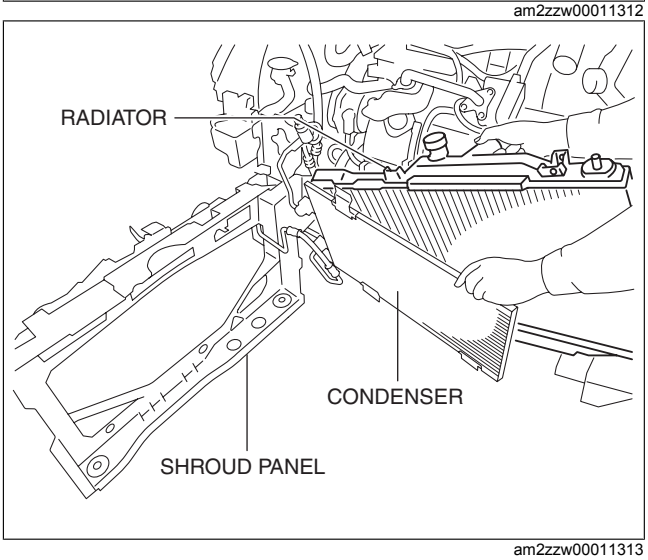
1. To prevent damage to the radiator core, protect it by covering the radiator surface with cardboard.

2. Remove the bolts securing the condenser and the cooler pipe to the radiator.



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3. Separate the radiator and the condenser, and remove the radiator.



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